Czech Technical University in Prague

FJFI - Department of Physics

02ELMA - Electricity and Magnetism

Makeup Exam

26.05.2025 / 13:30pm

Name					
Problem	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 90 minutes.
2. The maximum score for the test is 4 points and each question is worth 1 point.
3. Use of any class materials and calculators are not allowed.
4. Use only pens or non-erasable ink pens for the test; erasable pens are not allowed.
5. Write your answers legibly. Illegible answers will not be scored.
6. Write your answers in the provided space. If you need more space, additional sheets will be provided.

Question 1

A long, straight, solid cylindrical conductor of radius R carries a steady total current with homogenous current density J. Using Ampère's law, find the magnetic field $\vec{B}(s)$ (magnitude and direction) for

(a) $s < R$, — *0.5 pts*

(b) $s > R$. — *0.5 pts*

Solution:

Question 2

A long straight cylindrical region of radius R carries a spatially uniform time-varying magnetic field along the axis of the cylinder, given by

$$\vec{B}(t) = B_0 t^2 \hat{z}$$

for $r \leq R$ and $\vec{B}(t) = 0$ for $r > R$, where B_0 is a constant. Using Faraday's law in integral form

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A},$$

find the azimuthal electric field $\vec{E}_\phi(r, t)$ (magnitude and direction) for

(a) $r < R$, — *0.5 pts*

(b) $r > R$. — *0.5 pts*

Solution:

Question 3

The magnetic field component of an monochromatic electromagnetic wave propagating in free space is given by plane waves as

$$\vec{B}(z,t) = B_0 \sin(kz - \omega t)\hat{y}.$$

(a) What is the direction of propagation of the wave? — *0.1 pts*

(b) Find the electric field component of the wave. — *0.2 pts*

(c) Find the Poynting vector of the wave. — *0.2 pts*

(d) How much power is carried by the wave through a circular area of radius a in the plane perpendicular to the direction of propagation? — *0.2 pts*

(e) What is the direction of polarization of the wave? — *0.2 pts*

(f) If the wave goes through a polarizer oriented at an angle $\pi/4$ with respect to the direction of polarization, what would be the amplitude of the transmitted wave? — *0.1 pts*

Given: $\vec{B}_0 = \frac{1}{c}(\hat{z} \times \vec{E}_0)$ where c is the speed of light in vacuum.

Solution:

Question 4

A ring of radius R carries a total charge Q uniformly distributed over its circumference. The ring is free to rotate about its central axis. If the ring is spinning with an angular velocity ω find the following:

(a) The magnetic dipole moment $\vec{\mu}$ of the ring. — *0.5 pts*

(b) The spinning charged ring is then placed in a uniform magnetic field $\vec{B} = B\hat{z}$, where its axis of rotation is tilted at an angle θ with respect to the magnetic field direction. Find the torque $\vec{\tau}$ acting on the ring. — *0.5 pts*

Solution: